

PROJECT PROPOSAL - ECE 482/582, Fall 2003

PROJECT TITLE:

NAME: _____

PROJECT GOAL: What is it you are trying to do and what do you expect to learn from it?

Goal:

Learning expected:

SPECIFICATIONS: How will you know if you have accomplished the goal or not? Write up a list of "specifications" you plan to meet.

PROJECT PLAN: Describe your plan for the project - preliminary ideas sketches or layout, flow charts, circuit diagrams, etc.

EQUIPMENT/SUPPLIES: What will you need in the way of equipment, devices, and supplies in order to accomplish your goals?

Class Projects

Purpose:

The design project will let you design something exiting and educational using the knowledge you gain throughout the course. You are encouraged to think about new concepts or build on the concepts presented in the course. You may also make an experiment for the lab. Discuss this idea with the instructor. You should expect the project to be challenging and fun. You should also expect to spend a lot of time on it. This is where the learning occurs!

Some Project Ideas

Free-space LED communication system
Analog sending CD signal
Digital Communications

Fabry-Perot with curved mirrors, mode-matched to get TEM₀₀ mode

Look at diode laser with FPI. Requires special mirrors

Laser distance measurement device

Laser "eavesdropping" device to listen based on reflections of a window

Optical Second Harmonic Generation

Optical second harmonic generation seems like magic: focus a laser beam into a simple crystal, and out comes light at twice the frequency. The goal of this project is to develop a laboratory experiment demonstrating the concepts and techniques of second harmonic generation using a diode laser source. You will learn the basics of nonlinear optics.

Relative Intensity Noise vs. Bias Current and Temperature for a Semiconductor Laser

Laser Diode current controller

Differential Amplifier for photo-detection

Spectroscopy project

Optical tweezers

Fourier Optics Experiment

Be creative and come up with some of your own project ideas!